Universidad Central del Ecuador

Facultad de Filosofía Letras y Ciencias de la Educación

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Curso: 5to "B"

Fecha:

Tema: Vectores

① $\vec{A} = 3i + 4j$

$A = \sqrt{3^2 + 4^2}$

$A = \sqrt{9+16}$

$A = \sqrt{25} = [5m]$

$\tan\theta = \tan^{-1}\left(\frac{Ay}{Ax}\right)$

$\theta = \frac{4}{3}$

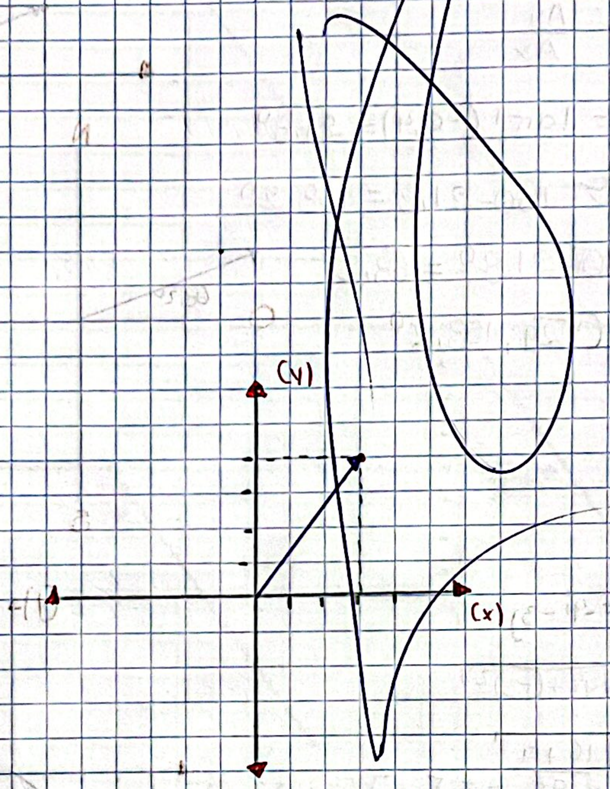
$\theta = \tan^{-1}(1{,}333) = 53.13°$

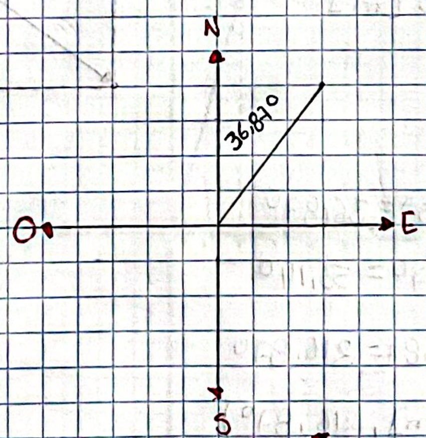
θ 180 - 53.13

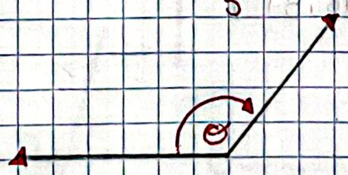
$\theta = 126{,}87$

90 - 53,13
= 36,87

$A = (5[m], 53,13]$







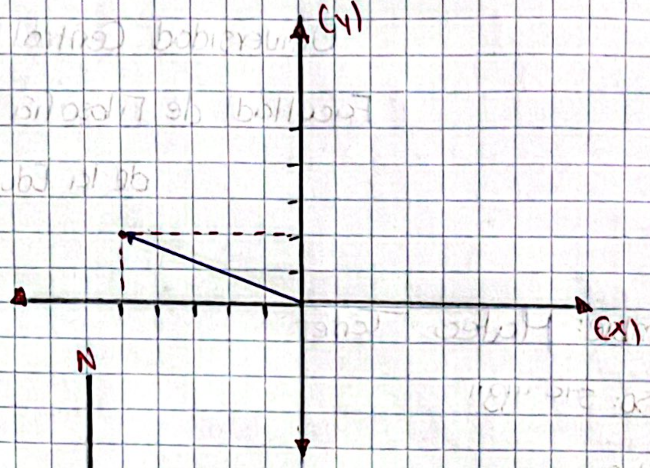
(2) $\vec{A} = -5\hat{i} + 2\hat{j}$

$A = \sqrt{(5)^2 + 2^2}$

$A = \sqrt{25 + 4} = \sqrt{29}$

$A = 5{,}38\ [m]$

$\tan\theta = \dfrac{A_y}{A_x}$

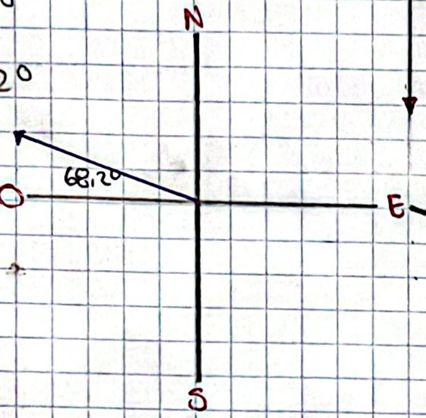


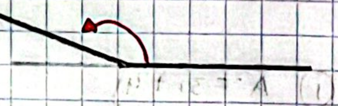
$\theta = \tan^{-1}(-0{,}4) = 21{,}8^\circ$

$\theta = 180 - 21{,}8 = 158{,}2^\circ$

$90 - 21{,}8^\circ = 68{,}2$

$A = (\sqrt{29},\ 158{,}2^\circ)$



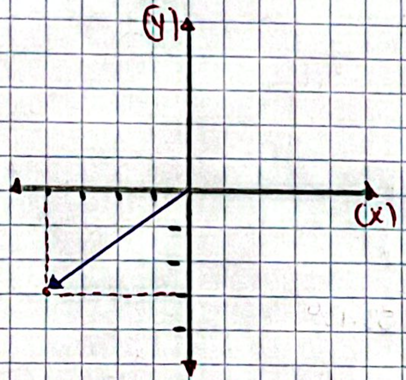


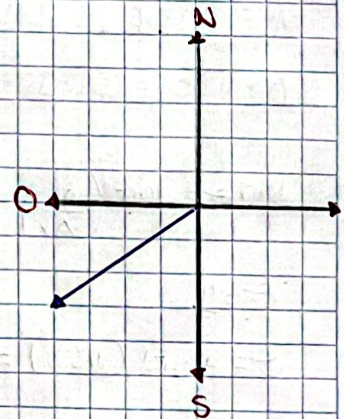
(3) $\vec{A} = -4\hat{i} - 3\hat{j}$

$A = \sqrt{4^2 + (-3)^2}$

$A = \sqrt{16 + 9}$

$A = \sqrt{25} = 5\ [m]$





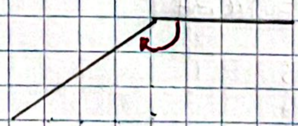
$\tan\theta = \dfrac{A_y}{A_x}$

$\theta = \tan^{-1}(0{,}75) = 36{,}87^\circ$

$90 - 36{,}87^\circ = 53{,}14^\circ$

$\theta = 180 + 36{,}87 = 216{,}87^\circ$

$A = (5\ [m],\ 216{,}87^\circ)$



④ $\vec{A} = 6\hat{i} - 8\hat{j}$

$A = \sqrt{6^2 + (8)^2}$

$A = \sqrt{36 + 64}$

$A = \sqrt{100}$

$A = 10\ [m]$

$\tan\theta = \frac{A_y}{A_x}$

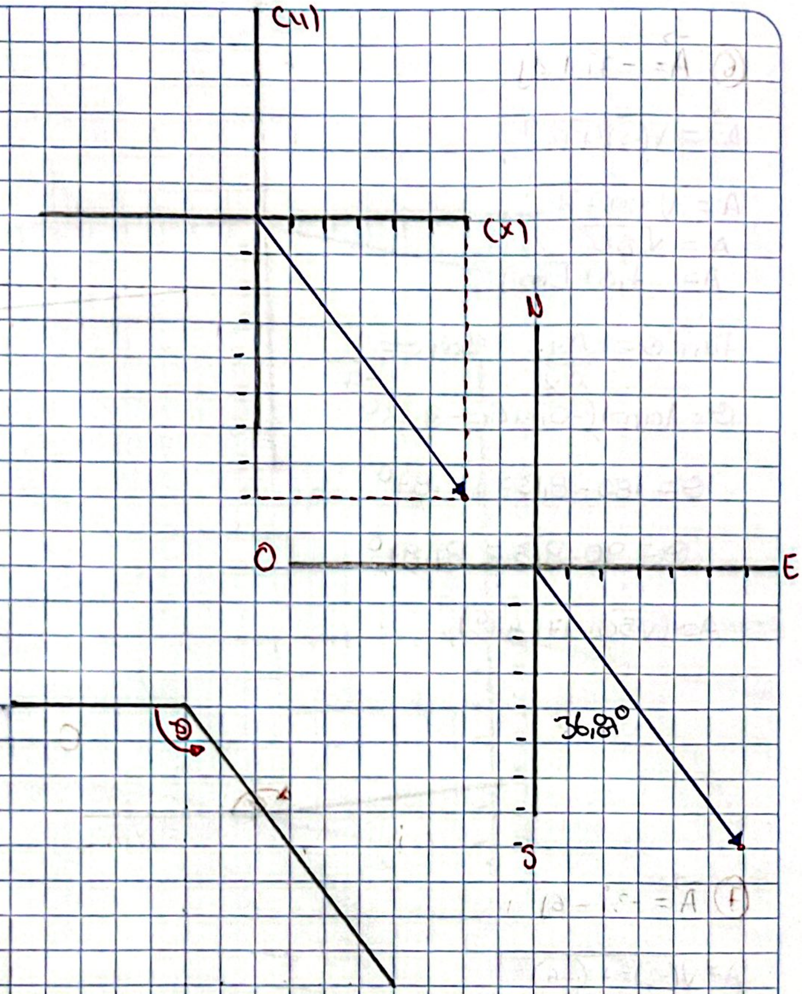
$\tan\theta = \frac{-8}{6}$

$\theta = \tan^{-1}(-1{,}33) = -53{,}13°$

$\theta = 180 - 53{,}13 = 126{,}87°$

$90 - 53{,}13 = 36{,}87°$

$A = (10[m], 126{,}87°)$



⑤ $A = 2i + 7j$

$A = \sqrt{2^2 + 7^2}$

$A = \sqrt{4 + 49}$

$A = \sqrt{53}$

$A = 7{,}28\ [m]$

$\tan\theta = \frac{A_y}{A_x}$

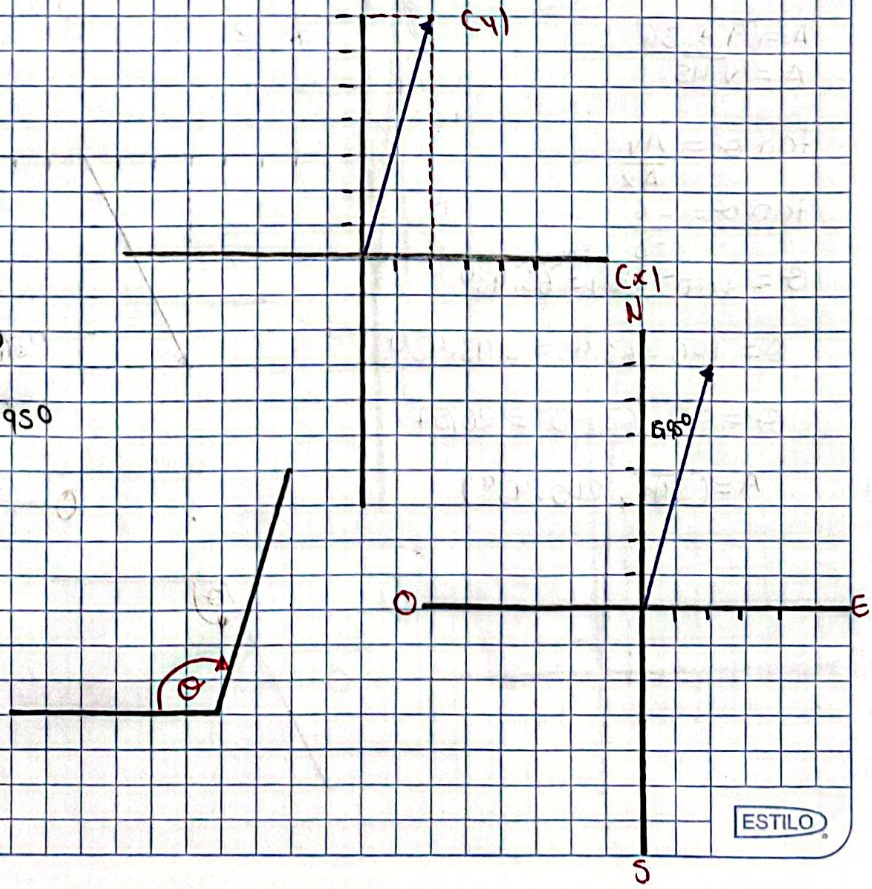
$\tan = \frac{7}{2}$

$\theta = \tan^{-1}(3{,}5) = 74{,}05°$

$\theta = 180 - 74{,}05 = 105{,}95°$

$\theta = 90 - 74{,}05 = 15{,}95$

$A = (\sqrt{53}, 74{,}05°)$



⑥ $\vec{A} = -7i + 1j$

$\vec{A} = \sqrt{(-7)^2 + 1^2}$

$A = \sqrt{49 + 1}$

$A = \sqrt{50}$

$A = 7{,}07$ [m]

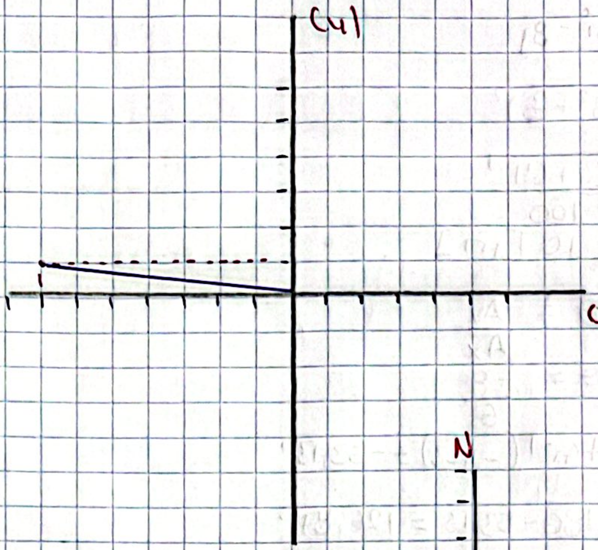
$\tan\theta = \frac{A_y}{A_x}$ $\tan\theta = \frac{1}{-7}$

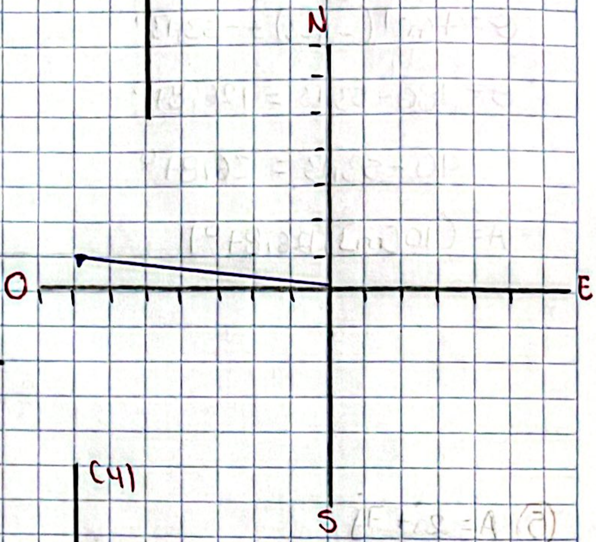
$\theta = \tan^{-1}(-0{,}142) = -8.13^\circ$

$\theta = 180 - 8{,}13 = 171{,}87^\circ$

$\theta = 90 - 8{,}13 = 81{,}87^\circ$

$A = (\sqrt{50}, 171{,}87^\circ)$





⑦ $\vec{A} = -3i - 6j$

$A = \sqrt{(-3)^2 + (-6)^2}$

$A = \sqrt{9 + 36}$

$A = \sqrt{45}$

$\tan\theta = \frac{A_y}{A_x}$

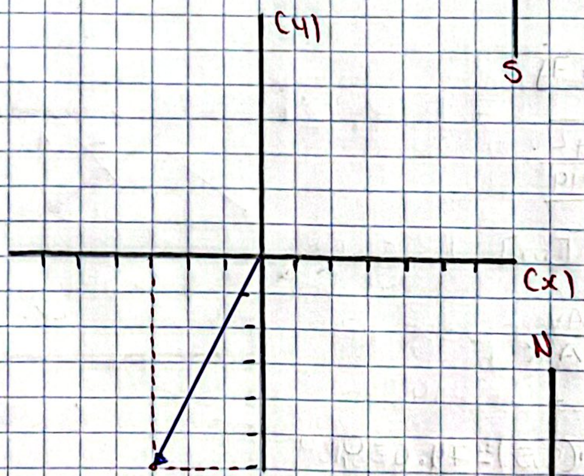
$\tan\theta = \frac{-6}{-3}$

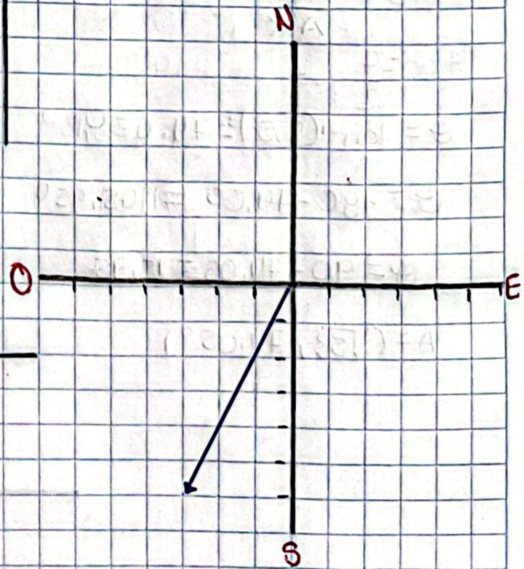
$\theta = \tan^{-1}(2) = 63{,}43^\circ$

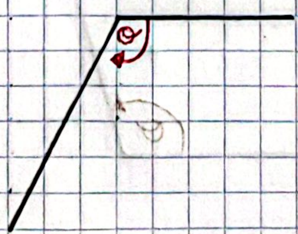
$\theta = 180 + 63{,}43 = 243{,}43^\circ$

$\theta = 90 - 63{,}43^\circ = 26{,}57$

$A = (\sqrt{45}, 245{,}43^\circ)$





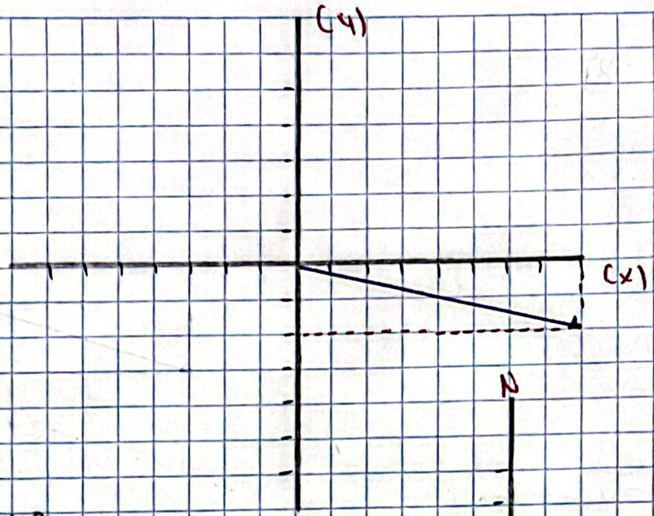


⑧ $\vec{A} = 8i - 2j$

$A = \sqrt{8^2 + (-2)^2}$

$A = \sqrt{64 + 4}$

$A = \sqrt{68}$



$\tan\theta = \frac{Ay}{Ax}$

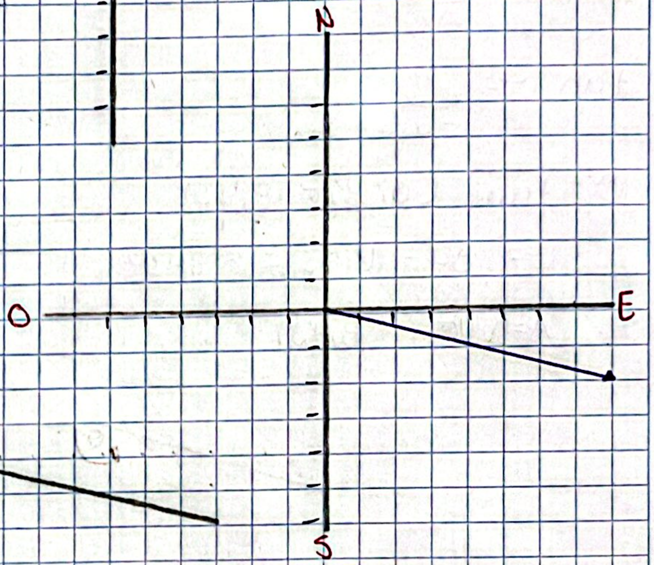
$\tan\theta = \frac{-2}{8}$

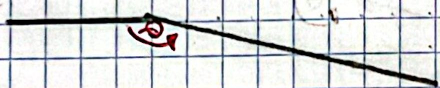
$\theta = \tan^{-1}(-0,25) = -14,04°$

$\theta = 180 - 14.04 = 165,96$

$\theta = 90 - 14,04 = 75,96$

$A = (\sqrt{68}, 165,96°)$



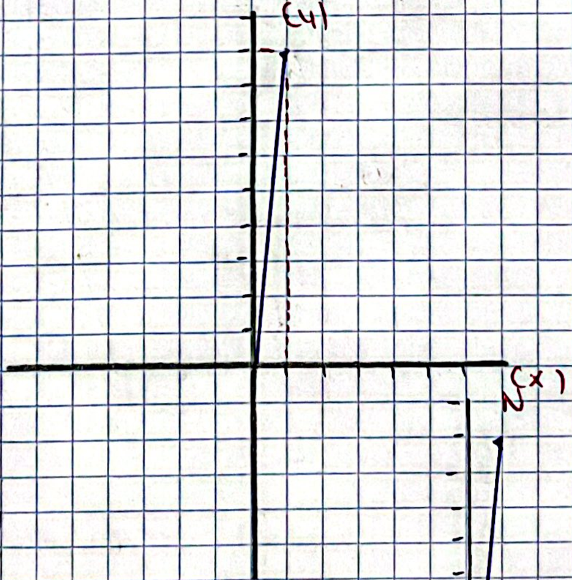


⑨ $\vec{A} = 1i + 9j$

$A = \sqrt{1^2 + 9^2}$

$A = \sqrt{1 + 81}$

$A = \sqrt{82}$



$\tan\theta = \frac{Ay}{Ax}$

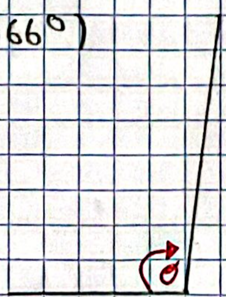
$\tan\theta = \frac{9}{1}$

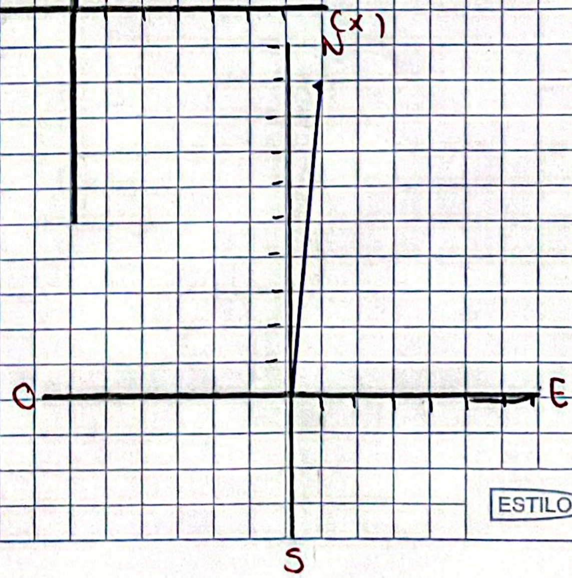
$\theta = \tan^{-1}(9) = 83,66°$

$\theta = 180 - 83,66 = 96,34$

$\theta = 90 - 83,66 = 6,34$

$A = (\sqrt{82}, 83,66°)$





ESTILO

(10) $\vec{A} = -6\hat{i} - 2\hat{j}$

$A = \sqrt{(-6)^2 + (-2)^2}$

$A = \sqrt{36+4}$

$A = \sqrt{40}$

$\tan\theta = \frac{A_y}{A_x}$

$\tan\theta = \frac{-2}{-6}$

$\theta = \tan^{-1}(0{,}33) = 18{,}43°$

$\theta = 180 + 18{,}43 = 198{,}43°$

$A = (\sqrt{40},\ 198{,}43)$